

OBSERVATIONS ON THE ETIOLOGY OF THE TOXEMIAS OF PREGNANCY.

THE RELATIONSHIP OF NUTRITIONAL DEFICIENCY, HYPOPROTEINEMIA, AND ELEVATED VENOUS PRESSURE TO WATER RETENTION IN PREGNANCY.*†

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APPROXIMATELY 25% of present-day maternal mortality is due to toxemia of pregnancy. Although such disorders as the anemias and polyneuritis of pregnancy have been included under this designation, these conditions may now be considered due to nutritional deficiency.¹⁻⁵ The term toxemia of pregnancy, as employed in this paper, designates the occurrence of arterial hypertension, with or without other abnormalities, during the latter months of pregnancy, in women known to have had normal blood pressures and no albuminuria early in gestation. The term eclampsia is reserved to designate the occurrence of convulsions and coma during the latter months of pregnancy, in labor, or during the puerperium, in women not presenting evidence of epilepsy, brain tumor or other adequate cause for the condition. Many observers believe that such events are the result of the action of one or more hypothetical toxins elaborated by the maternal or fetal organism. Although albuminuria is a common concomitant of toxemia, hypertension alone occurred in 585 of the 1036 cases of toxemia reported by Tillman and Watson.⁶

When convulsions and coma occur in the latter part of pregnancy or the puerperium, albuminuria and hypertension are usually present. However, convulsions and coma, considered to be eclampsia, have been reported without albuminuria by numerous clinicians,⁷⁻¹⁰ and more rarely, have been observed in the absence of arterial hypertension.¹⁰⁻¹¹

There remains, however, one other condition which is an exceedingly common concomitant of any and all of these other manifestations of toxemia, that is edema. In fact, most observers believe that either gross or occult edema is the *sine qua non* for the diagnosis of true toxemia of pregnancy. However, Fink¹² claims that 95% of all pregnant women have some degree of occult or gross edema.

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More than 60 years ago, Rosenstein¹³ advanced the thesis that cerebral edema was the cause of eclampsia. This he regarded as due to the effusion of serum from the "too-watery" blood into the cerebral tissues. Charpentier,⁸ 13 years later, agreed with several compatriots that a diminished amount of albumin in the blood was a predisposing cause in the production of eclampsia.

More recently Wieloch¹⁴ has observed the universal occurrence of water retention in his cases of toxemia of pregnancy, and Zange-meister¹⁵ has actually demonstrated edema of the brain by means of trephining the skulls of eclamptic patients. Prutz,¹⁶ and Arnold and Fay,¹⁷ among many others, have made similar observations postmortem. Bingham¹⁸ showed that toxemia developed only in those women who gained weight excessively. Although the cause of this weight gain was not determined, water retention might be its explanation. The cause of the marked water retention, which by general consensus of opinion¹⁹ is so uniformly present in toxemia, has been variously ascribed to an excess production of pituitary antidiuretic hormone,²⁰ to primary kidney disease,²¹ to a disturbance in the stability of the plasma colloids,²² and to pressure of the enlarged uterus on the ureters,²³ among many other hypothetic causes. However, no satisfactory evidence has been presented for any of these theories. No modern investigator has reexamined the hypothesis of Rosenstein, Prodhomme and others in the light of present-day knowledge concerning the mechanism of edema formation. Our understanding of this mechanism begins particularly with Starling's²⁴ demonstration of the rôle which the colloid osmotic pressure of the plasma proteins plays in counter-balancing the mechanical effect of the intra-capillary blood pressure. From Peters and Van Slyke's²⁵ excellent summary we learn that three main factors are involved: the intra-capillary blood pressure, the osmotic pressure of the plasma proteins, and the capillary permeability. In addition, edema apparently may occur as a result of the lack of certain still unknown accessory food factors (Yang and Huang,²⁶ Youmans,²⁷ Elsom²⁸ and others). Furthermore, Weech *et al.*²⁹ have recently shown that edema is formed in poorly nourished dogs at a much higher level of plasma proteins than in well-nourished dogs whose plasma proteins have been depleted by plasmapheresis. It is also apparent that edema cannot be formed unless there be supplied to the organism the necessary materials, that is, water and salt.

It is pertinent to consider at this point what changes occur in these various factors involved in water exchange during normal pregnancy. Grzechowiak³⁰ and Mufson³¹ have directly measured the intracapillary pressure in pregnancy. Although it is not elevated in normal pregnancy, they found it to be markedly so in eclampsia and many cases of toxemia. However, the work of Landis and Gibbon³² has so clearly demonstrated that the filtration of fluids out of the capillaries is directly proportional to the venous

pressure that studies of the latter in pregnancy by Runge³³ are important. The maximum venous pressures which he observed in normal non-pregnant subjects were 6.6 cm. and 7.2 cm. of water in the arm and leg respectively, contrasted to 8.9 cm. and 8.8 cm. respectively in normal women during the first 7 months of pregnancy and 9.8 cm. and 31.4 cm. respectively in normal women in the last 2 months of pregnancy. Barath and Weiner³⁴ found venous pressures of from 10 to 18 cm. of water in the arm at the end of normal pregnancy. It is thus apparent that normal pregnancy, particularly in the later months, may result in a distinct elevation of the venous pressure. If this is due to the mechanical effect of the enlarged uterus, one might expect it to be most marked in primiparæ with taut abdominal walls, in the obese, in the pyknic type, and in women with very large uteri from twin pregnancy or hydramnios. There is no doubt but that there is a higher incidence of toxemia in such individuals.^{35,36,37} It is also apparent that the venous pressure is tremendously raised during the active stage of labor, during which time contractions of the abdominal musculature and diaphragm result in expulsive efforts.³⁸ It may be significant that 53% of all reported cases of eclampsia summarized from every available source³⁹ occurred during labor and 21% shortly after delivery. Unless a mechanical effect is involved, it is difficult to understand why three-fourths of all cases of eclampsia should occur during this relatively short period of the gravid state.

The concentration of the plasma proteins has been regularly found lowered during pregnancy.^{22,41-44} This is corroborated by the data to follow. Eufinger,²² Plass,⁴³ Eastman⁴⁴ and others have found this lowering of the plasma protein concentration most marked in toxemias of pregnancy and particularly in eclampsia, which is also evident from the data in this paper. The albumin fraction which has 4 times the osmotic pressure of the globulin fraction is particularly reduced. Although the degree of reduction of the osmotic pressure of the plasma proteins in pregnancy is ordinarily not below the so-called critical level for edema formation in the normal individual, it should be remembered that, in the presence of a second disturbance affecting filtration, there may be a much higher critical level, as has recently been shown in heart disease by Thomson.⁴⁵

The cause of the lowering of the plasma protein concentration in pregnancy has not been clearly defined. One cause is apparent. It is recognized that the fetus draws upon the maternal organism for dietary factors, including its protein precursors, no matter what depletion of the mother results.¹ Therefore, if the mother's protein intake has been barely sufficient for her own economy and no change is made in her diet after pregnancy occurs, depletion of the mother's protein, especially in the last trimester of pregnancy, may be inevitable.

Disturbed gastro-intestinal function can condition dietary deficiencies. Such disturbances are common in pregnancy^{2,46,47} and could play a rôle in conditioning protein deficiency in certain pregnant women in spite of an apparently adequate diet. Definite evidence on this point is, however, lacking. The part played in pregnancy by a failure to manufacture proteins through a disturbance of the organs involved in their production is entirely hypothetical but possibly important.

Observations. *The Plasma Proteins in Normal Pregnancy and in Toxemia.* The concentration of the total plasma protein and of the plasma albumin was determined by a modification of the method of Howe⁴⁸ on oxalated venous blood drawn without stasis, after the subject had been recumbent for at least 15 minutes. These observations were made on blood from the antecubital veins of 65 pregnant women. Of these 35 were normal and in the last 2 months of pregnancy, 20 were suffering from toxemia during a similar period of pregnancy, and 10 had eclampsia. The 20 with toxemia were known to have had normal arterial blood pressures and no albuminuria early in pregnancy, although at least 3 had had toxemia in preceding pregnancies, 1 having had eclampsia twice. At the time that blood was withdrawn for these determinations, each of the 20 patients with non-convulsive toxemia had a blood pressure of 150 (systolic) and 100 (diastolic) or higher. Seven had from 0.2 to 2 gm. of albumin per liter of urine (Tsuchiya method). The remaining 13 had only very slight traces of albumin in the urine. Only 3 had gross edema.

Each of the 35 normal pregnant women was closely questioned as to her usual protein intake both during and prior to pregnancy. Before any chemical analysis was made of the plasma, each dietary history was evaluated as either average or low in protein content. Of these women 20 had had diets containing moderate amounts of protein (estimated at 60 to 100 gm. per day), and 15 had had definitely low protein diets (estimated at less than 60 gm. per day). The failure to encounter any women with high protein intakes can probably be explained by the fact that they were all clinic patients and hence unable to afford the more costly protein foods in large amounts.

In Chart I are presented the plasma albumin concentrations of the 65 women grouped according to the condition which they presented.

It is apparent that in toxemia, with or without convulsions, the plasma albumin concentration is decidedly lower than in normal pregnant women. The average concentration of the plasma albumin in the 10 patients with eclampsia was 2.6%, in the 20 patients with toxemia without convulsions 3.1% and in the 35 normal pregnant women 3.56%. The lowering of the plasma albumin concentration in the 20 patients with non-convulsive toxemia can hardly be

ascribed to a loss of albumin in the urine, as 13 of these women had only very slight albuminuria. The average concentration of plasma albumin was lower in the normal subjects with poor protein intakes than in those with average intakes.

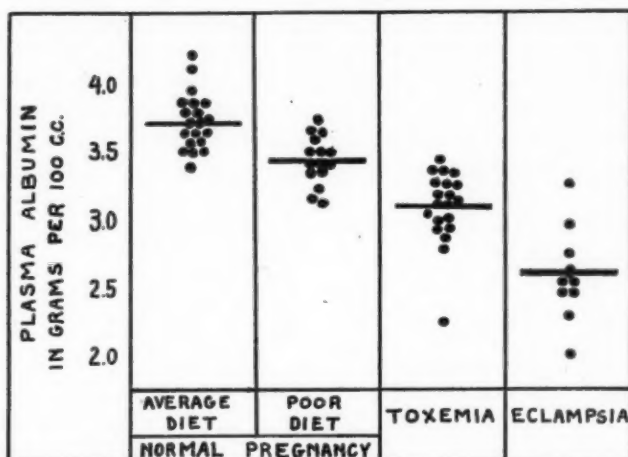


CHART I.—Plasma albumin concentration during the last 2 months of gestation in 20 normal women on average protein diets, in 15 normal women with low-protein intakes, in 20 women with non-convulsive toxemia of pregnancy, and in 10 with eclampsia. Each dot represents 1 patient. The horizontal lines represent the average values for each group.

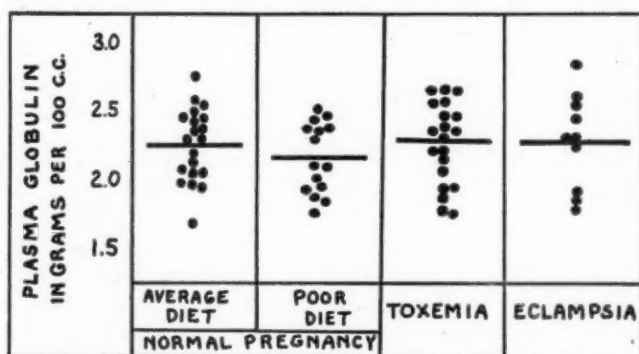


CHART II.—Plasma globulin concentration in the same group as in Chart I.

In Chart II the data on plasma globulin concentration obtained on the same 65 women is shown. The entire absence of correlation between diet, toxemia or eclampsia and the level of globulin is obvious.

Because, however, it is the colloid osmotic pressure exerted by both the albumin and globulin that is of importance with regard to

edema formation, the osmotic pressure of both fractions together has been calculated by Wells, Youmans and Miller's method⁴⁹ in each case (Chart III). The average osmotic pressure was lowest in eclampsia (175 mm. of water), next in non-convulsive toxemia (215 mm.), next in normal pregnant women with poor diets (232 mm.), and highest in those with good diets (258 mm.).

The Venous Pressure in Normal Pregnancy and in Toxemia. The venous pressure of 20 pregnant women during the last trimester of pregnancy, whose arterial blood pressures were 150 mm. Hg systolic and 100 mm. Hg diastolic or higher, and of 20 normal pregnant women during the last 2 months of gestation was determined by

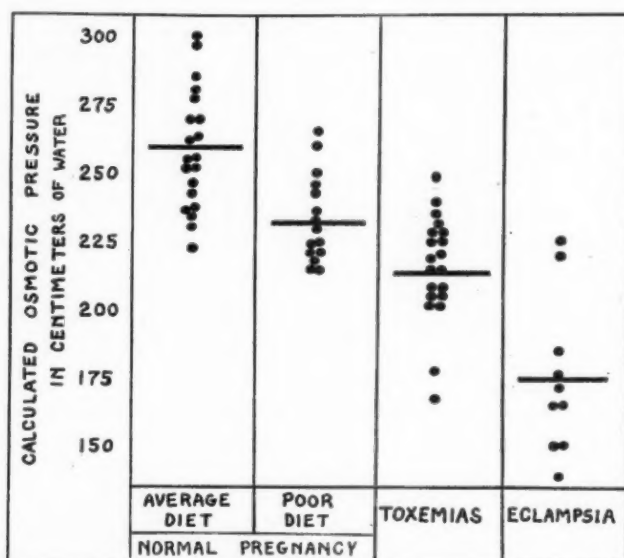


CHART III.—Calculated osmotic pressure of the total plasma proteins in the same group as in Chart I.

the method of Moritz and Tabora.⁵⁰ Readings were made after the subjects had been in a recumbent position for at least 15 minutes. The average venous pressure of normal non-pregnant individuals has been given by Eyster⁵¹ as 5 cm. of water, with a maximum of 11 cm. During the last 2 months of normal pregnancy (Chart IV) the average venous pressure of 20 women was 10 cm. of water with a maximum of 18 cm. The average venous pressure of the 20 patients with toxemia was 13.3 cm. of water, a value not significantly different from normal pregnant women but distinctly higher than in normal non-pregnant subjects.

The impression obtained was that the venous pressure was particularly elevated in the short, stocky, obese women and in the

extremely thin and rather frail appearing women whose abdomens seemed hardly large enough to accommodate a full-term pregnant uterus.

Dietary Histories of Women with Toxemia of Pregnancy. The dietary histories of 20 pregnant women with toxemia, were carefully studied. Six of these women had had severe nausea and vomiting for several months early in pregnancy which had greatly limited their total intake, concentrated carbohydrates being their chief food during the period of emesis. Two other patients had had

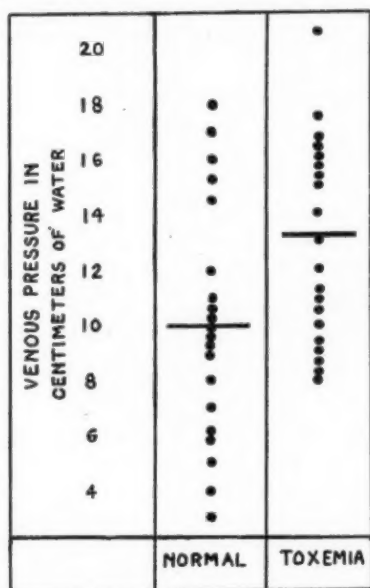


CHART IV.—Venous pressures in 20 normal pregnant women and in 20 with non-convulsive toxemia of pregnancy, all within the last 2 months of gestation. Dots and horizontal lines as before.

moderate emesis which had caused them to restrict their food to a certain extent. Of the 20 women, 18 had eaten little meat or other protein foods, not only during pregnancy but often over a period of years. The daily protein intake of these patients was estimated at less than 50 gm. daily. At least half of them stated that they were "poor meat eaters." Of the 20, 9 specified that they were very fond of and ate sweets and pastries in large amounts. Fifteen, who had accurate knowledge of their weight before pregnancy, had gained 20 to 44 pounds during gestation.

Results of Treatment of Toxemia of Pregnancy with a Special Diet High in Protein. Of the 20 pregnant women with toxemia, each of whom had an arterial blood pressure of 150 (systolic) and 100 (diastolic) or higher, 15 were given a diet containing 260 gm. of protein, 150 gm. of carbohydrate and 70 gm. of fat (2270 calories).

The protein was supplied in the form of lean meats, egg whites, skimmed milk and 300 gm. of raw liver pulp daily. Fluids were allowed freely. No purgation was employed. These patients were in the hospital during the period of observation but were not confined to bed. Ten of these 15 also received by daily intramuscular injections 5 cc. of a vitamin B₁ concentrate and 5 cc. of Solution Liver Extract—Lilly (N. N. R.).* The latter is believed to contain the pellagra and black-tongue preventative fraction of the vitamin B complex.

The weight changes of these 15 women at the end of a fortnight of this treatment are shown in Chart V. In general, weight was

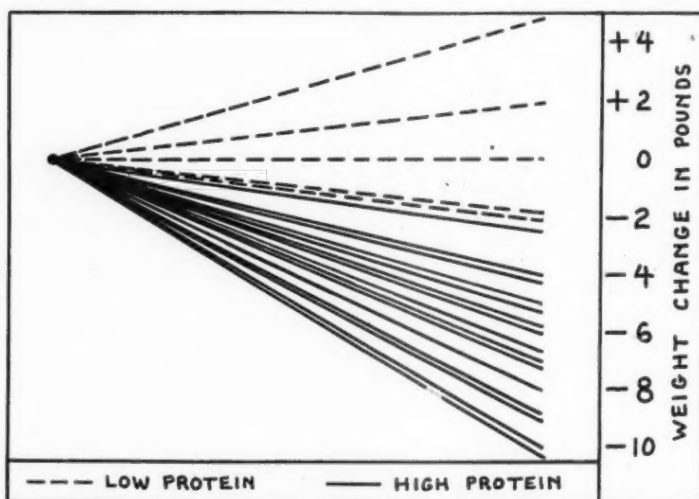


CHART V.—Weight changes after 2 weeks of 5 women with non-convulsive toxemia of pregnancy receiving a 20 gm. protein diet and of 15 similar women receiving a 260 gm. protein diet and parenteral injections of vitamin B.

lost steadily during this period in spite of the fact that they had daily about 2270 calories and were largely at rest. The amount of weight lost depended particularly on the proportional amount gained during the earlier months of pregnancy. In 3 patients gross edema completely disappeared, so that there was a marked loss in weight within 20 days. After the period of weight loss or within 3 weeks or less, the patients usually showed slow gains in weight consistent with the caloric intake. Rarely did the weight reach a level as high as that observed upon admission to the hospital. That this loss of weight during the first 2 or 3 weeks of treatment was due to the disappearance of either gross or occult edema seems to be very probable.

* Kindly supplied by Eli Lilly and Co., Indianapolis, Ind.

Coincident with the loss of weight, symptoms such as headache and visual disturbances were abated. In Chart VI, the arterial blood pressure of these women is shown at weekly intervals. In no case did the pressure rise after treatment was instituted. It is of interest that in every case the arterial blood pressure fell to below a systolic level of 140 mm. of mercury and to below a diastolic level of 90 mm. of mercury within 3 weeks of commencing treatment. The records indicate that the more acute the onset of the arterial

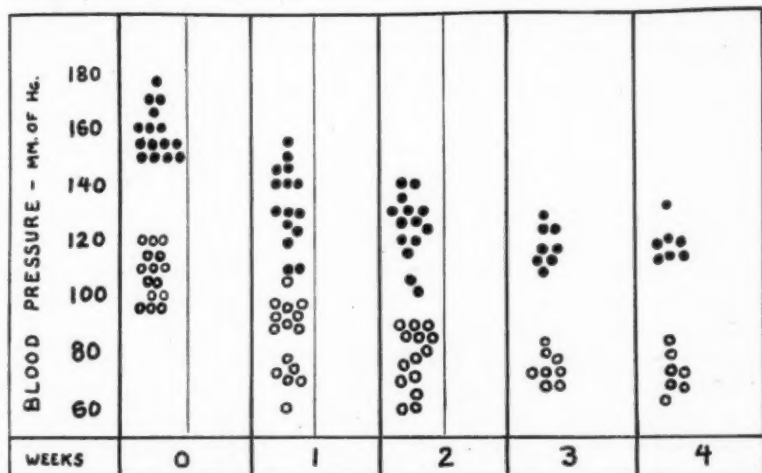


CHART VI.—Arterial blood pressures in successive weeks of 15 women with non-convulsive toxemia of pregnancy receiving a high-protein diet. The solid circles represent the systolic blood pressure, the open circles, the diastolic blood pressure. Seven of the patients were delivered by the end of the third week. No postpartum pressures are shown.

hypertension and the earlier the treatment, the more rapid was the response.

Albuminuria either remained unchanged or decreased in degree. No evidence suggesting that such a diet might lead to uremia can be drawn from Table 1, in which are presented the non-protein nitrogen values for these 15 women before and after the institution of the high-protein diet.

The calculated osmotic pressure of the plasma proteins in these women rose in 2 weeks an average of 7% of the initial level, usually by virtue of rises in both albumin and globulin fractions.

No fetal mortality occurred after the institution of this special diet.

After 3 of the above patients received the special diet in the hospital for 1 month, they returned to their homes, continuing the diet to the best of their abilities, during a period of 1 to 2 months

TABLE 1.—THE NON-PROTEIN NITROGEN OF THE BLOOD PLASMA IN 15 PATIENTS WITH NON-CONVULSIVE TOXEMIA OF PREGNANCY BEFORE AND AFTER ADMINISTRATION OF A HIGH-PROTEIN DIET.

Before mg. per 100 cc.	After mg. per 100 cc.
15	25
20	27
18	25
18	30
18	33
22	32
13	35
17	25
15	27
18	23
18	31
13	25
14	28
16	27
11	22
Average 16.4	Average 27.6

before each was delivered. In no instance was there any return of toxemic symptoms. Weight curves showed no evidence of recurrence of occult edema.

Results of Treatment of Toxemia of Pregnancy with a Low-protein Diet. Five of the 20 pregnant women with toxemia, with as nearly as possible identical signs and symptoms as the 15 women mentioned immediately above, were given a diet containing 20 gm. of protein, 65 gm. of fat and 400 gm. of carbohydrate. This diet contained 2265 calories or approximately the same number as the high-protein diet. However, it contained no meat, milk, fish, or eggs. The sodium chlorid content was essentially the same as that of the high-protein diet. The vitamin content was low. Fluids were allowed freely and no purgation was employed. The weights of these 5 women during a period of 2 weeks are shown in Chart V. One patient gained 5.25 pounds, a second 1.75 pounds, and the remaining 3 lost 0.25, 1.75 and 2 pounds each. The blood pressure (Chart VII) in 2 cases had fallen to normal within 2 weeks, in the remaining 3 patients it either rose or was stationary. Albuminuria was unchanged in 3 of the 5 patients, and increased from a faint trace to 0.3 and 15 gm. per liter respectively in the remaining 2 women. The calculated osmotic pressure of the plasma proteins fell an average of 9% of the initial value in these 5 women, the loss being chiefly at the expense of the albumin fraction.

Discussion. It is apparent that edema disease in pregnancy, mechanically complicated as it is by the large intraabdominal mass of the pregnant uterus, an elevated venous pressure, and the constant drain on the protein reserves of the mother, must necessarily differ from edema disease in the non-pregnant individual. It is of

interest, however, that Yang and Huang²⁶ have recorded the occurrence of convulsions as well as arterial hypertension in the "wet" form of beriberi. Cowgill and his associates⁵² have shown that the need for vitamin B increases with elevation of the metabolism. Since the basal metabolic rate is raised about 20% in pregnancy, considerably more vitamin B is probably required by pregnant women. Hence a deficiency of this factor may easily occur in pregnancy.

There is evidence to suggest that hemoglobin regeneration following the administration of iron is slower in pregnant than in non-pregnant patients with hypochromic anemia.¹ One would expect

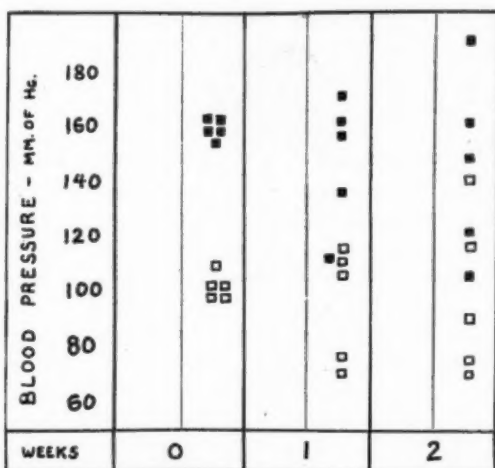


CHART VII.—Arterial blood pressures in successive weeks of 5 women with non-convulsive toxemia of pregnancy receiving a low-protein diet. The solid squares represent the systolic blood pressure, the open squares, the diastolic blood pressure.

similarly that regeneration of plasma protein would be slower in pregnant women because of the constant demands of the fetus.

The weight changes of the 15 patients receiving the high-protein diet may be ascribed to any number of factors, such as rest in the hospital, or an increased nitrogen excretion resulting in diuresis. However, the failure of the control group to show changes of a similar magnitude suggests that the rest obtained is but a minor factor. Whether the patients who received the high-protein diet and injections benefited because of the protein content of the diet or because of receiving accessory food factors or both, cannot be determined at present.

The rôle of tissue resistance, lymph flow, and capillary permeability in edema formation in pregnancy has not been touched upon because of the paucity of our knowledge of these factors. The one

pertinent study⁵³ which has been made of edema fluid in toxemias of pregnancy has shown its protein content to be low, suggesting that there is no abnormal alteration of capillary permeability such as exists in certain other types of edema.

Because the arterial blood pressure is subject to spontaneous fluctuations both in normal and hypertensive subjects, and because rest frequently lowers both systolic and diastolic levels, no conclusions should be drawn from the fact that there was a lowering of the arterial blood pressure in each of the 15 women treated with the high-protein diet. It should be noted that this also occurred in 2 of the 5 patients receiving the low-protein diet.

The observations recorded in this paper fail to substantiate the prevalent belief that patients with toxemia of pregnancy should be treated by means of a restricted protein intake. It must be remembered, however, that none of these patients had either albuminuria or arterial hypertension during the first half of pregnancy. Approximately 5 of every 6 patients diagnosed as having toxemia of pregnancy in our pre-natal clinic were excluded from this study because there was no satisfactory evidence that the blood pressure was normal and the urine free of albumin during the early months of pregnancy. It might be added that a few observations, not recorded here, fail to show that any harm results from the use of high-protein diets in pregnant women with pre-existing arterial hypertension or albuminuria, or does such a diet appear to have any beneficial effect on such women.

Summary. 1. In 20 cases of non-convulsive toxemia of pregnancy the average osmotic pressure of the plasma proteins was 215 mm. of water, in 10 cases of eclampsia it was 175 mm. of water and in 20 normal pregnant women with adequate diets it was 258 mm. of water. In 15 normal pregnant women who had partaken of diets low in protein the average osmotic pressure of the plasma proteins was 232 mm. of water.

2. The average venous pressure in 20 normal pregnant women during the last 2 months of pregnancy was 10 cm. of water (range 3 to 18 cm.) and in 20 patients with non-convulsive toxemia of pregnancy 13.3 cm. (range 8 to 20.5 cm.), values which are at least twice as high as average figures for normal non-pregnant subjects.

3. The dietary histories of 20 toxemic pregnant women showed a low-protein intake, probably less than 50 gm. per day, frequently over a period of years.

4. Fifteen women suffering from toxemia of pregnancy were treated by means of a diet containing 260 gm. of protein, 150 gm. of carbohydrate and 70 gm. of fat. Ten also received parenteral injections of a vitamin B₁ concentrate and Solution Liver Extract—Lilly (N.N.R.), the latter because of its content of the pellagra-preventative portion of the vitamin B complex. Each of the 15 women showed a loss of weight readily attributable to decrease of

occult or gross edema, together with a gradual disappearance of the signs and symptoms of toxemia. No fetal mortality occurred after treatment had been instituted.

5. Five similar pregnant women suffering from toxemia were treated by means of a diet of approximately equal caloric value containing 20 gm. of protein, 400 gm. of carbohydrate, and 65 gm. of fat. No significant loss of gross or occult edema occurred. Two of the patients became decidedly worse during a period of 2 weeks.

6. It is believed that a manifestation of toxemia of pregnancy is water retention. This water retention probably occurs as a result, among other factors, of a lowered osmotic pressure of the plasma proteins, usually in the presence of an increased venous pressure. The results recorded in this paper suggest that a restricted dietary intake of protein in pregnancy is harmful, and that no injurious consequences follow the administration of high-protein diets to women with toxemia of pregnancy. The beneficial results observed in these patients may well have been due to the large protein intake and to the parenteral administration of accessory nutritional factors.

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(Titles have been omitted for sake of brevity).

ERYTHEMA ANNULARE RHEUMATICUM.

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IN 1922 Lehnendorff and Leiner,¹ under the title "Erythema Annulare" published an account of a typical exanthem occurring during the course of rheumatic endocarditis. Lehnendorff,² in 1930, again described "Erythema Annulare Rheumaticum" and emphasized the fact that it occurs only in rheumatic children with endocardial involvement. While making the rounds at the Mautner-Markhof Kinderspital in Vienna in 1925, the late Professor Leiner demonstrated this rash to me in a child with rheumatic endocarditis. Since then I have had the opportunity of following a group of children suffering with rheumatic endocarditis at the Spaulding School for Crippled Children, and during the course of my examinations I have noted this distinctive rash in 6 children. The following cases may be cited: